



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Adaptive Software Engineering – (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	MedFlow AI – An Agentic Cloud-Native Healthcare Intelligence Platform using Microservices, DevOps and DevSecOps	
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Abstract

Healthcare organizations increasingly rely on digital systems to manage patients, clinicians, appointments, medical records, laboratory services, pharmacy operations, billing, and emergency care, yet many conventional healthcare management applications remain fragmented and process-centric, requiring extensive manual coordination across departments while providing limited intelligent support, resulting in duplicated information, delayed communication, inefficient appointment and queue management, difficulty interpreting medical documents, fragmented patient records, and slower responses to operational and emergency needs. **MedFlow AI** addresses these challenges through a service-oriented microservices architecture integrated with artificial intelligence and modern software engineering practices, providing role-based workflows for patients, doctors, nurses, receptionists, laboratory staff, pharmacists, billing personnel, emergency staff, and administrators, with core capabilities including authentication, patient profiles and medical histories, appointment management, queue handling, medical records, laboratory reports, prescriptions, pharmacy operations, billing, notifications, and emergency support. An AI service built using Spring AI and the Google Gemini API coordinates six specialized agents—Patient Assistant, Doctor Assistant, Emergency Assistant, Pharmacy Assistant, Billing Assistant, and Hospital Operations Assistant—capable of multi-step reasoning and domain-specific tool use rather than single-shot question answering, supporting medical-report summarization, clinical information organization, appointment and queue guidance, medication and inventory insights, bill explanations, and operational recommendations, while functioning strictly as decision-support tools under human oversight rather than autonomous diagnostic or treatment systems. The system is implemented using **Java 21, Spring Boot, Spring Cloud, React.js, and PostgreSQL** with a database-per-service architecture, while **Spring Security/JWT, API Gateway, Eureka, and OpenFeign** provide security, service discovery, API routing, and inter-service communication. **Docker, Kubernetes, and AWS** support cloud-native deployment, while **Jira, GitHub Actions, SonarQube, Trivy, OWASP ZAP, Prometheus, and Grafana** enable Agile development, CI/CD, DevSecOps practices, security and code-quality validation, and system monitoring, collectively providing a scalable, secure, reliable, and extensible healthcare platform that demonstrates the integration of agentic AI with modern cloud-native software engineering.

Signature of the Guide

HOD